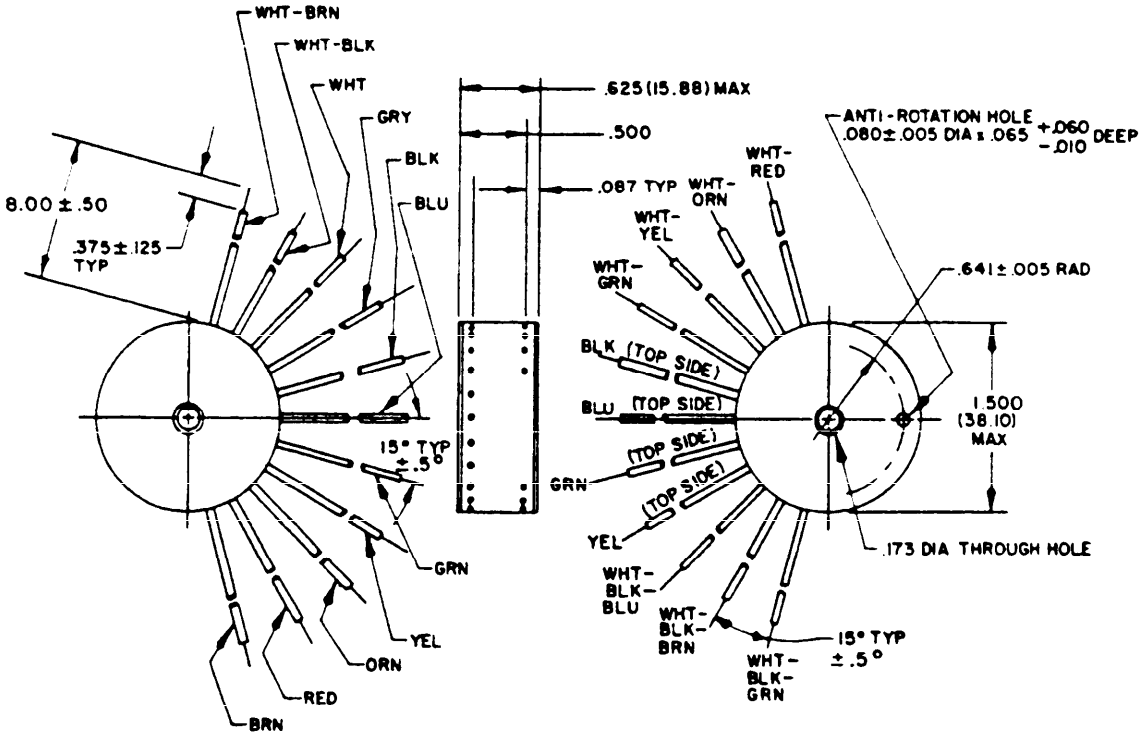
MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEPDOWN AND STEP-UP, TF5S03ZZ900

This specification is approved for use by all Departments and Agencies of the Department of Defense.

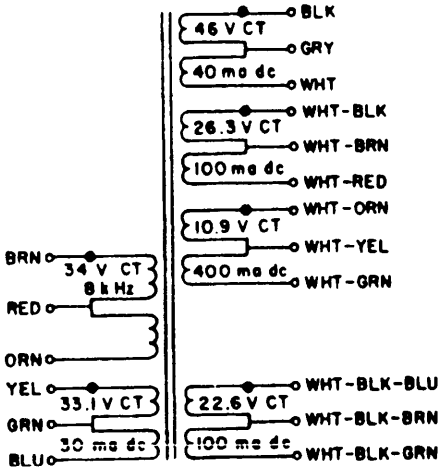
The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of Specification MIL-T-27.



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FSC 5950

CIRCUIT DIAGRAM
AND MARKING



WV: 50 V MAX
ALTITUDE: 70, 000 FT

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Leads under YEL, GRN, BLU & BLK omitted.
4. Marking shall be on the side of the case.
5. Metric equivalents are in parentheses.
6. Unless otherwise specified, tolerances are ±.005 (.13 mm) for three place dimensions.

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REQUIREMENTS: (When color abbreviations in parentheses, i.e., (BRN-ORN) are used, they indicate the extreme terminals of the winding.)

Electrical ratings:
- Primary voltage (BRN-ORN): 34 V ct at 8 kHz, square wave.
- Primary power: 13.5 VA.
- Primary current: 0.397 ampere rms.
- Secondary voltage:
  - (YEL-BLU): 33.1 V ct at 30 mAdc ±1%.
  - (BLK-WHT): 46 V ct at 40 mAdc ±1%.
  - (WHT/BLK-WHT/RED): 26.3 V ct at 100 mAdc ±1%.
  - (WHT/ORN-WHT/GRN): 10.9 V ct at 400 mAdc ±1%.
  - (WHT/BLK/BLU-WHT/BLK/GRN): 22.6 V ct at 100 mAdc ±1%.
- Working voltage:
  - (BRN-ORN): 50 V maximum.
  - (YEL-WHT/BLK/GRN): 50 V maximum.
- DC resistance at 25°C:
  - (BRN-ORN): 0.580 ohm maximum.
  - (YEL-BLU): 4.60 ohms maximum.
  - (BLK-WHT): 5.46 ohms maximum.
  - (WHT/BLK-WHT/RED): 1.40 ohms maximum.
  - (WHT/ORN-WHT/GRN): 0.25 ohm maximum.
  - (WHT/BLK/BLU-WHT/BLK/GRN): 1.30 ohms maximum.

Design and construction:
- Dimensions and configuration: See figure.
- Duty cycle: Continuous.
- Case: Encapsulated.
  - Material: Epoxy.
- Altitude: 70,000 feet maximum.
- Terminals: Special, type E, stranded, plastic-coated (PTFE), #24 AWG, 19 strands of #36 in accordance with MIL-W-16878.
  - Terminal length: 8 inches ±0.5.
- Weight: 2.2 ounces.
- Operating temperature range: -55° to 130°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage:
- At sea level: 100 Vrms.
- At barometric pressure: 100 Vrms.

Electrical characteristics:
- No load: With 40 Vrms, 8000 Hz square wave across (BRN-ORN), current in (BRN-ORN), shall not exceed 17 milliamperes rms.
- Full-load output at 25°C: With 34 V, 8 kHz, square wave across (BRN-ORN), the voltage across:
  - (YEL-BLU): Shall be 33.1 V ±1%, 0.03 ampere.
  - (BLK-GRY) or (GRY-WHT): Shall be 23 V ±1%, 0.04 ampere.
  - (WHT/BLK-WHT/BRN) or (WHT/BRN-WHT/RED): Shall be 13.15 V ±1%, 0.10 ampere.
  - (WHT/ORN-WHT/YEL) or (WHT/YEL-WHT/GRN): Shall be 5.45 V ±1%, 0.40 ampere.
  - (WHT/BLK/BLU-WHT/BLK/BRN) or (WHT/BLK/BRN-WHT/BLK/GRN): Shall be 11.30 V ±1%, 0.10 ampere.
- Polarity: Additive, with terminals ORN, YEL; BLU, BLK; WHT, WHT/BLK: WHT/RED, WHT/ORN: WHT/GRN and WHT/BLK/BLU connected.

Temperature rise: 30°C maximum, with 34 V, 8 kHz across (BRN-ORN) at an ambient temperature of 100°C maximum.

Voltage regulation: 1.5% maximum.

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Marking location: See figure.

Part number: M27/83-01.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI, SG
Navy - EC, SH, OS
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - MU, WC, ME
Navy - MC, AS, SH, OS
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5950-0506-3)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-T-27/83 TRANSFORMER, POWER, STEPDOWN AND STEP-UP, TF5S03ZZ900

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

To detach this form, along this line

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